

Ling Ling HE

Aspiring Computer Science & AI Engineer

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PROFILE

Computer Science & AI student with a strong analytical foundation in algorithms, data structures, and system design. Experienced in building high-performance systems, including embedded controls and real-time navigation algorithms. Proficient in Java and Python with practical expertise in algorithm optimisation and sensor fusion. Passionate about developing scalable solutions and leveraging advanced analytics to solve complex technical challenges in collaborative team environments.

EDUCATION

BSc Computer Science and Artificial Intelligence | Queen Mary University of London Sep 2024 – Present
Expected graduation: **2027**

- **AI modules:** Introduction to Robotics, Introduction to Artificial Intelligence, Generative Deep Learning, AI for Decision Making
- **Programming modules:** Algorithms and Data Structures, Procedural Programming, Object-Oriented Programming, Fundamentals of Web Technology
- **Systems modules:** Computer Systems and Networks, Logic and Discrete Structures, Information Systems Analysis, Operating Systems
- Module Engagement Achievement - Fundamentals of Web Technology
- Machine Learning Society: Team-based Kaggle Competitions and machine learning tutorials

Uxbridge College

Sep 2021 - Jun 2024

- A Levels: Mathematics, Psychology, and Sociology
- GCSEs: Mathematics (9), English (4), Biology, Media, and Sociology

PROJECTS

Minecraft Settlement Procedural Generation – Ongoing

Mobile Robot Assembly & Navigation - Introduction to Robotics

- Assembled a four-wheel mobile robot, integrating motors, sensors, and an Arduino microcontroller
- Implemented navigation algorithms in Python (Arduino IDE) for real-world environments, including obstacle avoidance, line and wall following, bug algorithms, and goal tracking
- Tested, tuned and validated robot performance in physical and Tinkercad simulation environments
- Achieved 0-1 cm distance measurement error within a 0-40 cm range through systematic ultrasonic sensor testing
- Integrated hardware and embedded software to achieve autonomous robot behaviour
- Documented a technical report covering robot assembly, navigation algorithms, and experimental results

Horse Racing Simulator - Object-Oriented Programming

- Developed a horse racing simulator using Java with a focus on Object-Oriented Programming principles, including encapsulation, inheritance, and polymorphism
- Designed and implemented an interactive GUI using Swing, allowing users to start races, view results, and track horse statistics
- Improved understanding of event-driven programming

Personal Portfolio Website - Fundamentals of Web Development

- Designed and built a responsive personal portfolio website using HTML, CSS, JavaScript, and PHP
- Implemented mobile-friendly layouts for consistent performance across desktop and mobile devices
- Developed a PHP backend to handle server-side processing and form submissions
- Gained practical experience in full-stack development and responsive UI design

WORK EXPERIENCE

Customer Assistance | T4

Mar 2025

- Prioritised tasks and collaborated with team members during peak periods to support efficient operations, timely order fulfilment, and positive customer experiences
- Prepared and served bubble tea orders for in-store customers and Uber Eats deliveries, providing efficient, customer-focused service in a fast-paced environment
- Maintained a clean and organised workstation in compliance with hygiene, health, and safety standards

COMMUNITY ENGAGEMENT & RETAIL VOLUNTEERING

Event Photographer | Chinese New Year & Lantern Festival

Feb-Mar 2026

- Captured high-quality event photography featured on official organiser media channels and People's Daily Online

Thames Hospice

Feb 2024

- Delivered exceptional customer service by welcoming and assisting customers professionally
- Applied verbal communication and interpersonal skills to engage the public, distribute informational leaflets, and promote the charity's mission to support fundraising efforts

Barnardo's

Feb 2023

- Maintained coordinated product displays to improve visual presentation and accessibility
- Facilitated smooth shop operations by assisting with merchandising, restocking, and general store upkeep

British Heart Foundation

Feb 2022

- Strengthened communication and interpersonal skills by engaging with customers, addressing customer inquiries and needs
- Managed product labelling and organisation to improve in-store presentation

SKILLS

Programming languages: **Java, Python**

Web Technologies: **HTML, CSS, JavaScript, PHP**

Robotics & Embedded Systems: **Arduino, Mobile Robotics, Manipulator Robot Kinematics**

Core concepts: OOP, data structures, machine learning fundamentals

Tools: **Git/GitHub, Linux, VS Code, Google Colab**

Languages: Fluent English, Native Chinese (Mandarin)